Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

DIFFERENTIAL AND INTEGRAL CALCULUS 1 - 104003

FINAL B
September 14th, 2022

Your ID Number:

Guidelines

1. **Duration: 3 hours**.
2. This exam has 4 problems for a total of 100 points.
3. Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed. You will be provided with a dictionary (by demand) on the examination class (by one of the proctors).
4. Write your answers clearly, completely and concisely. Quote any results that you use. No credit will be given to answers which are not justified.

Problem 1 (40 points)

Let
$$f(x) = \frac{x^2 - 49}{x^2 + 5x - 14}.$$

1. Give the natural domain of f.
2. Find the derivatives f' and f''.
3. Find the critical points of f and identify the function's behavior at each one.
4. Find where the curve is increasing and where it is decreasing.
5. Find the points of inflection and determine the concavity of the curve.
6. Find horizontal and vertical asymptotes. Give the range of f.
7. Sketch the curve $y = f(x)$ together with any asymptotes that exist.

Problem 2 (25 points)

Evaluate the following integrals.

1. $\int_0^{\pi/4} \frac{1+\sin x}{\cos^2 x} dx$.
2. $\int x^3 \ln x \, dx$.
3. $\int \sin^2 2x \cos^3 2x \, dx$.

Problem 3 (20 points)

Evaluate the following limits.

1. $\lim_{x\to\infty} \frac{\sqrt{9x+1}}{\sqrt{x+1}}$.
2. $\lim_{x\to 0^+} \ln x - \ln(\sin x)$.
3. $\lim_{x\to -\frac{\pi}{3}} \frac{3x+\pi}{\sin(x+\frac{\pi}{3})}$.

Problem 4 (15 points)

For each of the following series decide if they converge or diverge. If they converge find their sum.

1. $\sum_{n=1}^{\infty} \frac{2}{10^n}$.
2. $\sum_{n=1}^{\infty} \sqrt{\frac{n+1}{n^2+2}}$.